

Topology First-Year Exam, May 2017, Part 1

Work the following problems and show all work. Support all statements to the best of your ability.

1. Prove that the circle S^1 is not homeomorphic to the closed 2-disk B^2 .
2. Let $A \subset \mathbb{R}^2$ be an infinite countable subspace.
 - (a) Can A be connected?
 - (b) Is $\mathbb{R}^2 \setminus A$ connected?
3. Let $f : X \rightarrow \mathbb{R}^2$ be continuous injective map. Is f an embedding if
 - (a) $X = (0, 1)$?
 - (b) $X = [0, 1]$?
 - (c) $X = [0, 1)$?
4. Show that a continuous injective map $f : (0, 1) \rightarrow \mathbb{R}$ is an embedding.
5. Show that every compact Hausdorff space is normal.

Answer the following with complete definitions or statements or short proofs.

6. Describe all connected subsets of the real line $\mathbb{R}$.
7. State the Contraction Mapping Theorem.
8. Let $\mathbb{R}_\ell$ denote the reals with the lower limit topology, i.e. topology defined by the basis $\{[a, b) \mid a, b \in \mathbb{R}\}$.
Is $\mathbb{R}_\ell$
 - (a) connected?
 - (b) regular?
9. State the Baire Category Theorem.
10. Show that a connected metric space cannot be infinite countable.
11. State the Urysohn Lemma.
12. Give definition of a quotient map.
13. Are the spaces $(0, 1) \times [0, 1)$ and $[0, 1) \times [0, 1)$ homeomorphic?
14. Describe the one-point compactification of $X = (0, 1) \cup (2, 3)$.
15. Let X be a locally connected space. Is every component of X closed?